

Embrace the Unknown: Russell's Paradox and The Halting Problem

↪ MA274 Final Presentation ↪

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1. Russell's Paradox
2. The Halting Problem
3. 🤪🤪🤪 They're the same?
4. Math is broken $\setminus (\circ \blacksquare \circ |||)/$
5. Math is fixed $\setminus (\cdot \forall \cdot)/$ (mostly $(\setminus (\circ \blacksquare \circ |||)/))$

Let's entertain a self-containing set, does such a set exists?

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Fibonacci sequence:

$$f(n) = f(n - 1) + f(n - 2)$$

Hilbert's Curve

$$H(n) = (H(n-1) \curvearrowright \pi/2, \quad H(n-1), \quad H(n-1) \curvearrowleft \pi/2)$$

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What's the beef with sets?

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- **A set** = a collection of objects
 - $\{\text{😺}, \text{😼}\}, \mathbb{R}, \mathcal{P}(\mathbb{R}), \dots$
- Sets can contain other sets as elements.
- Question: can a set contain itself?
- Tempting idea: consider the set of all sets that do not contain themselves...

Is there a barber in town who shaves everyone that doesn't shave themselves?

The Russell Set. Suppose a set of all sets exists...

Let $R = \{X \in \text{SETS} : X \notin X\}$.

So $X \in R \iff X \notin X$

Consider $R \in R$?

Case 1 $R \in R$. Then $R \notin R \quad \hookrightarrow$

Case 2 $R \notin R$. Then $R \in R \quad \hookrightarrow$

If you build your math on axiomatic set theory, and the set theory falls apart, your entire math falls apart...

$$0 = \emptyset$$

$$1 = \{\emptyset\}$$

$$2 = \{\emptyset, \{\emptyset\}\}$$

$$3 = \{\emptyset, \{\emptyset, \{\emptyset\}\}\}$$

⋮

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$\vdots$

fixed

*"Hardly anything more unfortunate can befall a scientific writer than to have one of **the foundations of his edifice shaken after the work is finished**. This was the position I was placed in by a letter of Mr. Bertrand Russell, just when the printing of this volume was nearing its completion." - Gottlob Frege, Grundgesetze Vol. 2*

The infinite loop is a common bug. Can I write a program to detect if a given program and input will continue to run forever or not (halt)?

The infinite loop is a common bug. Can I write a program to detect if a given program and input will continue to run forever or not (halt)?

Proposition: No

Suppose such a program exists, call it `checkhalt` :

```
def checkhalt(another_program):  
    # black magic to check ✨  
    if the_program_will_halt:  
        return True  
    return False
```

Then, define another program, call it `annoying` :

```
def annoying(another_program):  
    if checkhalt(another_program(another_program))  
        while True: pass # loops infinitely  
    return # halts
```

```
def annoying(another_program):  
    if checkhalt(another_program(another_program)): # 🖱️🖱️🖱️🖱️  
        while True: pass # loops infinitely  
    return # halts
```

Finally, consider `annoying^2 = annoying(annoying)`.

Case 1 `annoying^2` loops infinitely.

Then `checkhalt(annoying^2)` is `True`. Which means `annoying^2` halts.

Case 2 `annoying^2` halts.

Then `checkhalt(annoying^2)` is `False`. Which means `annoying^2` loops.



SETS

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............. Turns out, we CAN'T assume SETS is a set.

But what other kind of "collection" do we have?

- **Extensionality:** $A = B \iff A \subset B^B \subset A$
- **Power set:** $\forall A, \exists \mathcal{P}(A)$

Comprehension: If S is a set and $P(x)$ is a predicate of set theory, then

$$\{x \in S : P(x)\}$$

is also a set.

This kills Russell's Paradox.

Take the L like a champ. It is not the end of the world if your math fails. You're (you Frege) a pretty cool guy.

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No self-referencing sets!!!!